SEPCiN Offshore Areas of Interest Radar-based Surveillance Mapping for Early Oil (Spill) Anomaly Detection.

Issued by Geomatics Department: 2nd, January 2020
Date of Image Mapping: 1st, January 2020 T17:48:16 UTC
Author: Geomatics Department, Oyetunde Simisola
Urgency – ~~High~~/~~Medium~~/Low

Summary: Satellite surveillance tasking for **1st, January 2020 T17:48:16 UTC**

Surveillance tasking for **1st, January 2020 T17:48:16 UTC** identified two low confidence anomalies in close proximity to the Bonga FPSO approximately 0.8km² and 2.2km² in area respectively.

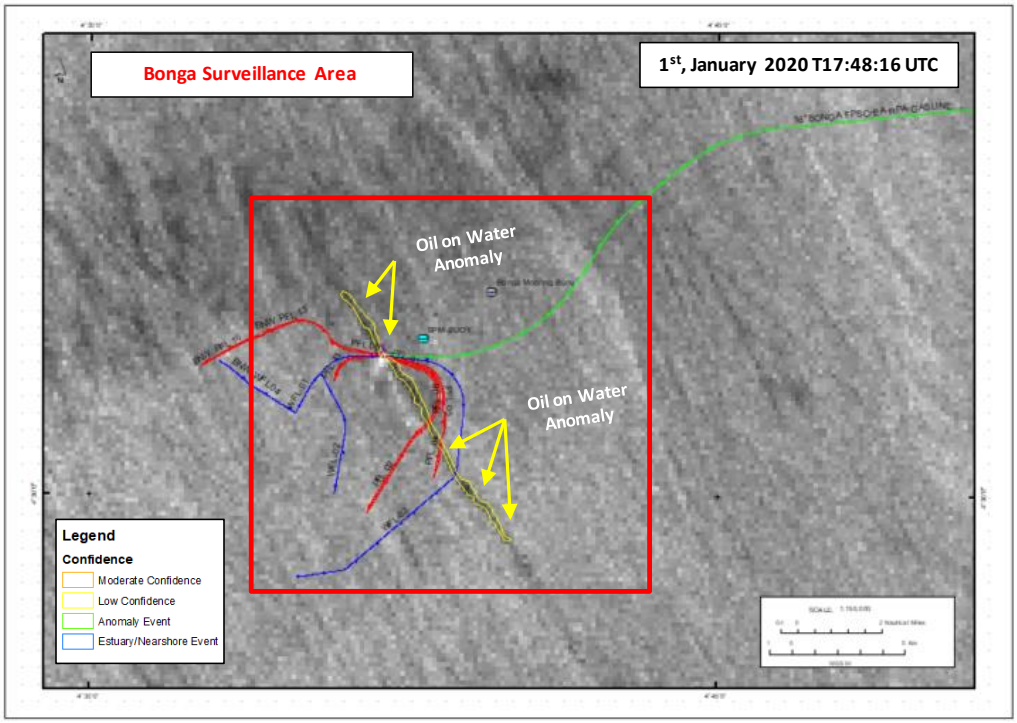
Recall that surveillance tasking for **17th November,2019 T05:22:47 UTC, 3rd November 2019 T05:31:00 UTC,7th October, 2019 T17:56:40, 6th July,2019 T05:31:06UTC, 23rd July,2019 T05:35:18UTC, 8th August,2019 T17:48:26UTC, 20th August,2019 T17:56:42UTC and 23rd August,2019 T05:31:03UTC** also identified three low confidence anomaly events in close proximity to Bonga FPSO **(See next slides).**

Several anomaly events have been identified around the Bonga area over the last several weeks which may all be related.

Overflight and/or In-situ monitoring by on site personnel is recommended

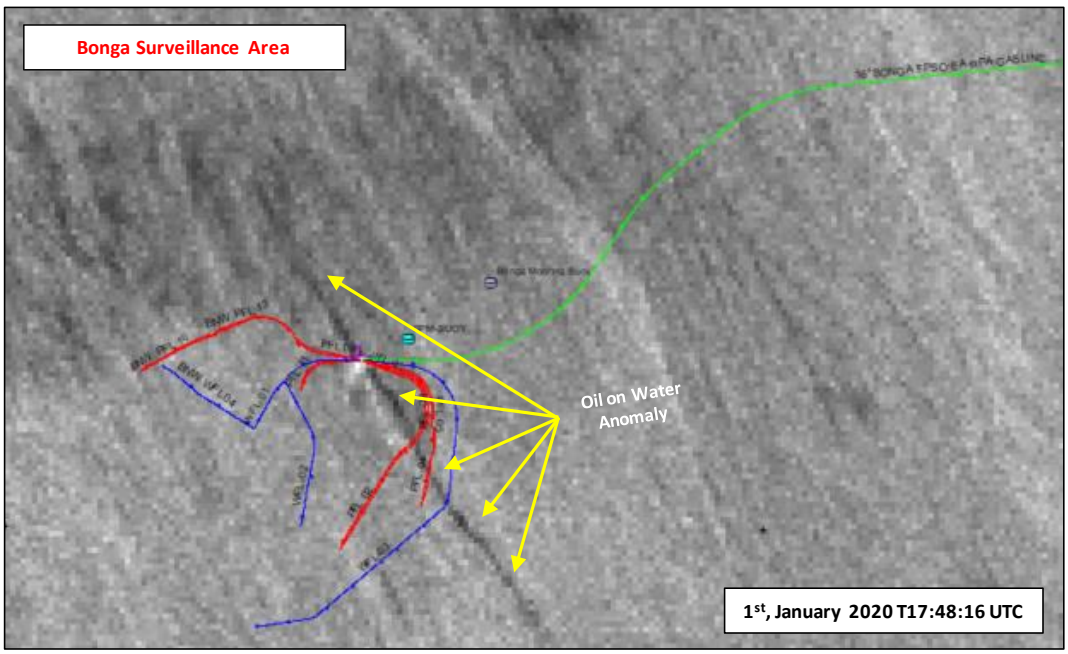
The next satellite pass over the area is planned for **4th, January 2020 5:23:49AM UTC.**

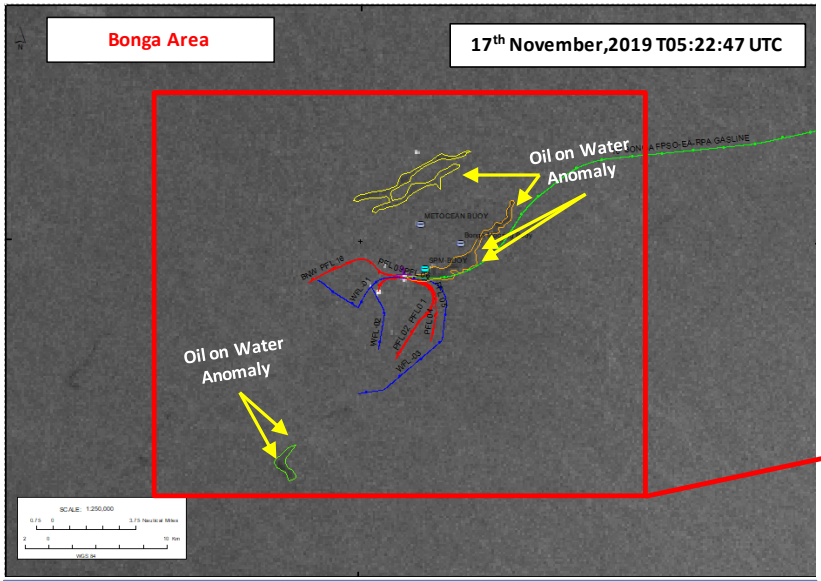
For further information: Evarie Peter, Simisola Oyetunde, Chituru Obowu or Chuks Bisike-Ojiako

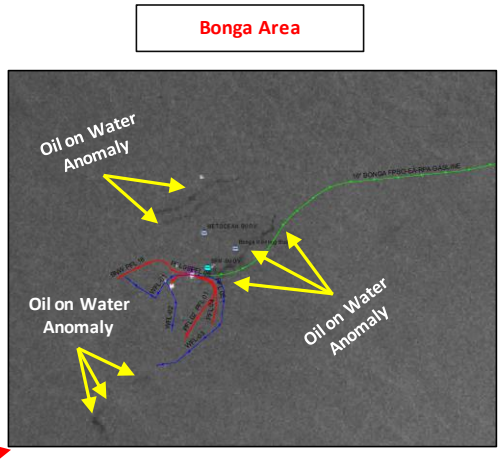


Oil Analysis from MDA, Radarsat 2, SCNB Beam, VV polarization **17th November,2019 T05:22:47 UTC**

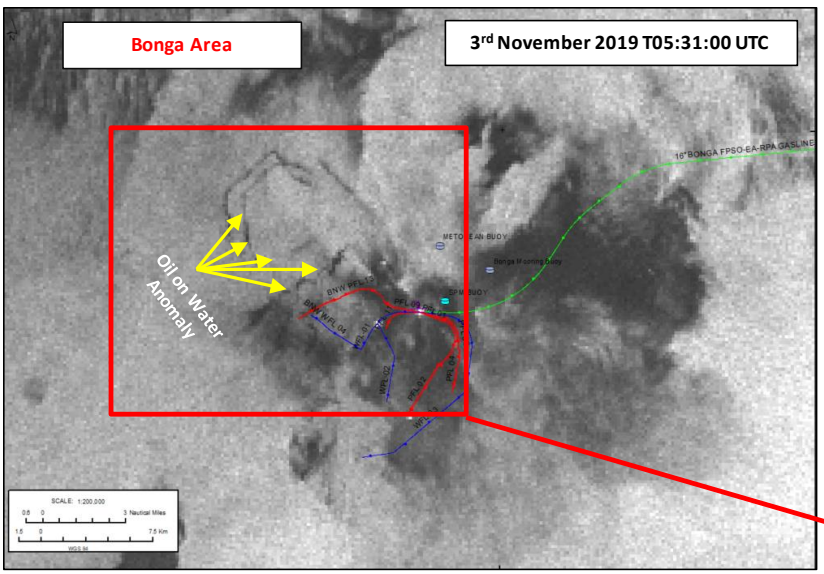
- One medium and two low confidence anomaly around the Bonga FPSO area approximately 7.8km²,7.7km²,2.05km² in area respectively

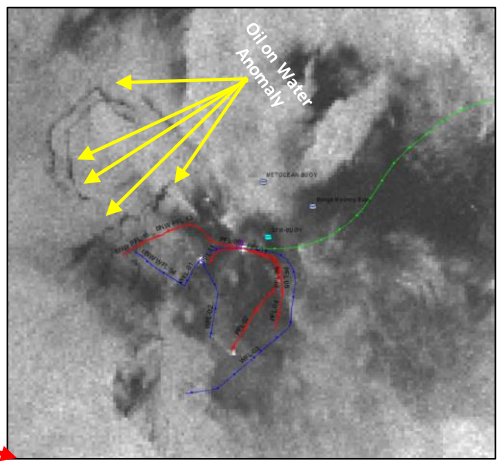






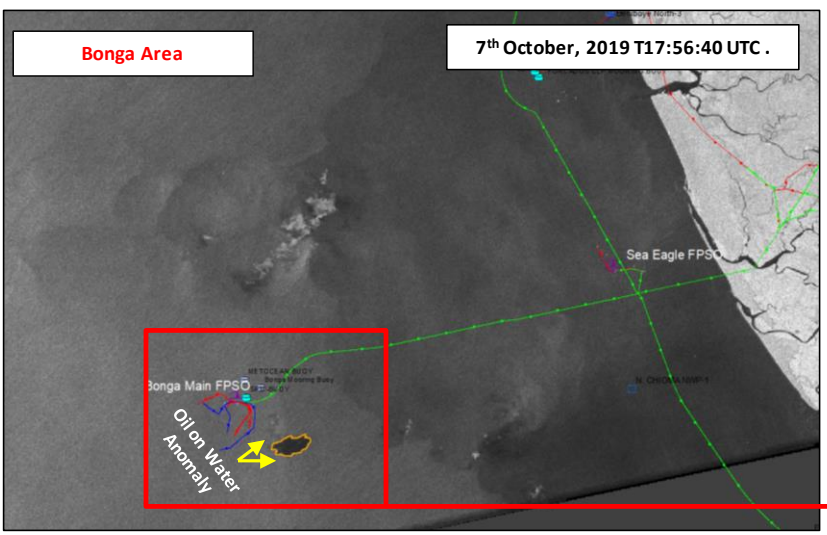
Oil Analysis from MDA, Radarsat 2, SCNB Beam, VV polarization, **17th November,2019 T05:22:47 UTC** identified one medium and two low confidence anomaly around the Bonga FPSO area approximately 7.8km², 7.7km², 2.05km² in area respectively

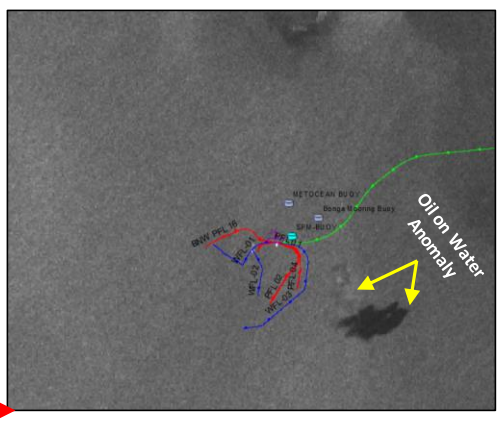




Oil Analysis from MDA, Radarsat 2, SCNA Beam, VV polarization, **3rd November 2019 T05:31:00 UTC**

- Surveillance tasking for **3rd November 2019 T05:31:00 UTC** identified low confidence anomalies approximately 8.5km², 2.3km² and 1.6km² respectively in area and 13.79KmNW, 9.03KmNW and 6.46KmNW respectively in close proximity to Bonga FPSO





Oil Analysis from MDA, Radarsat 2, SCNB Beam, VV polarization. **7th October, 2019 T17:56:40 UTC .**

- One moderate confidence anomaly event ~10km north east of the Bonga FPSO.

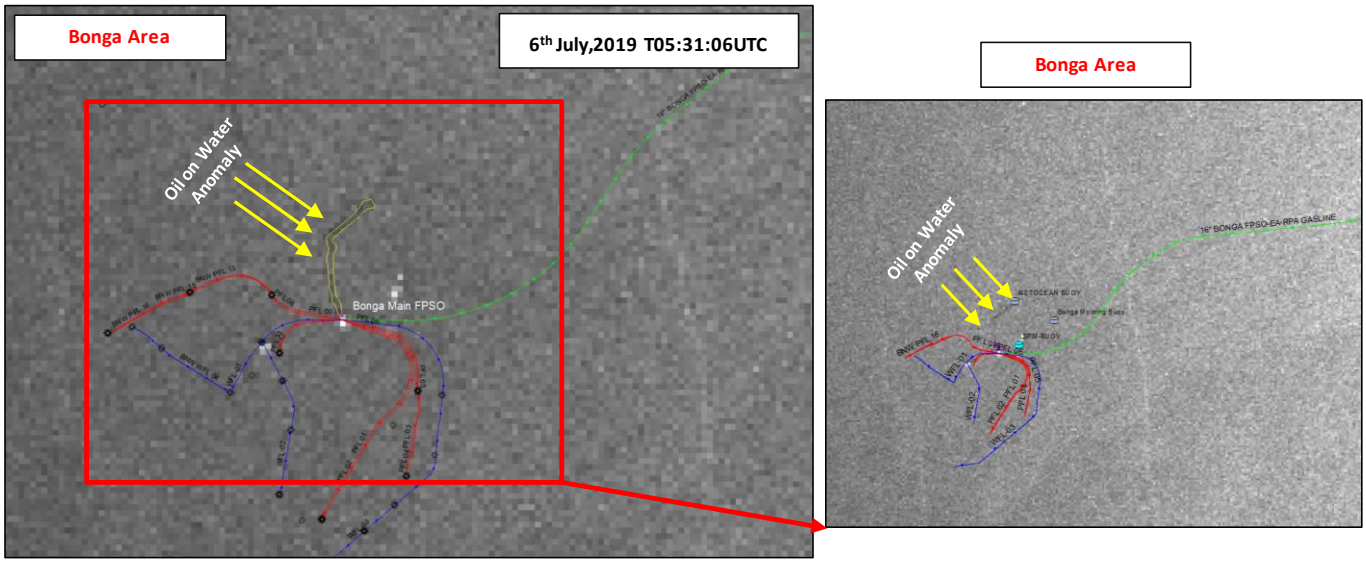


Image 6th July,2019 T05:31:06UTC

- Oil Analysis from MDA, Radarsat 2, SCNA Beam, VV polarization.A low confidence anomaly identified approximately 4.6km in length in close proximity to Bonga FPSO.

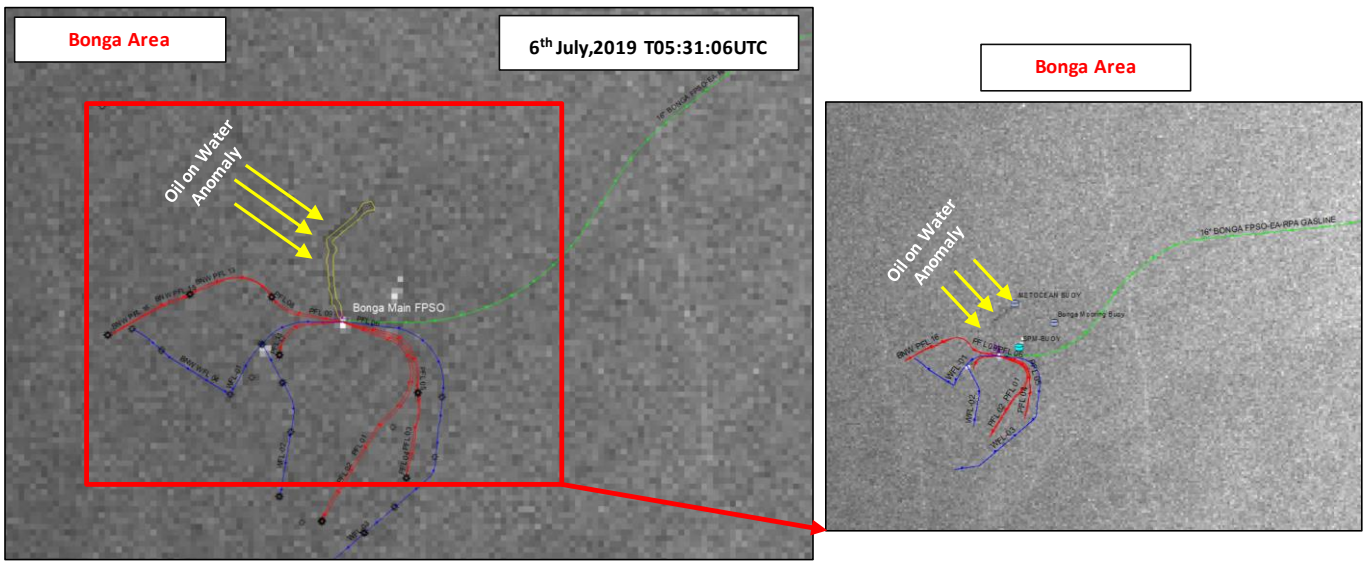


Image 6th July,2019 T05:31:06UTC

- Oil Analysis from MDA, Radarsat 2, SCNA Beam, VV polarization.A low confidence anomaly identified approximately 4.6km in length in close proximity to Bonga FPSO.

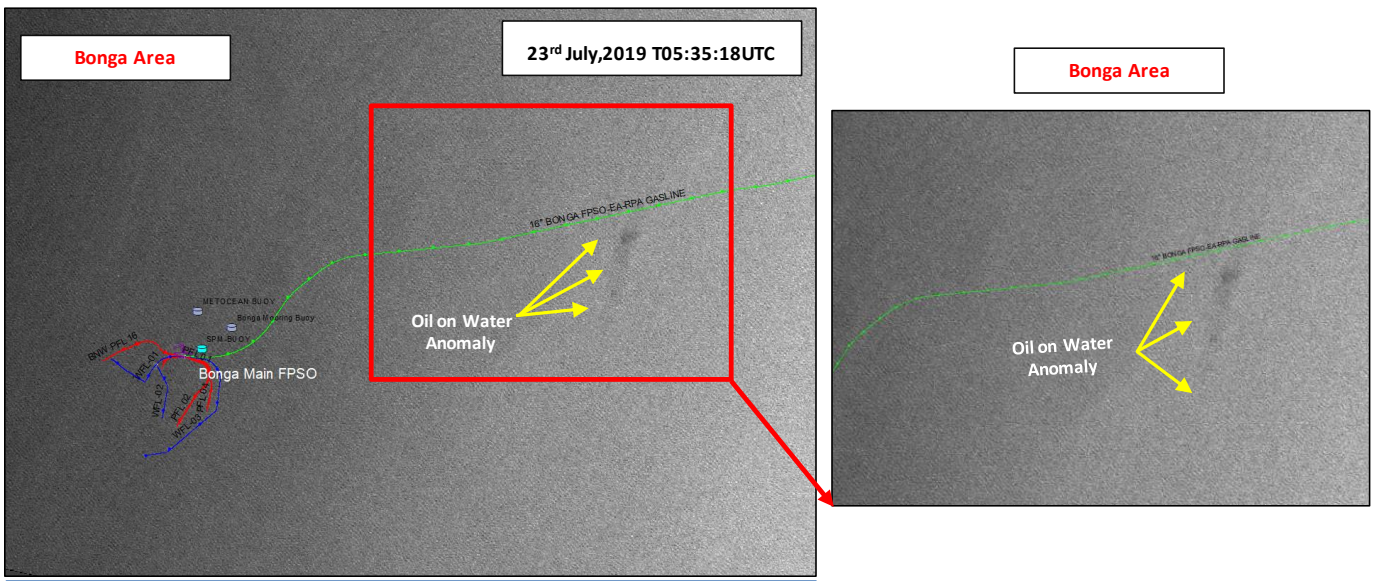
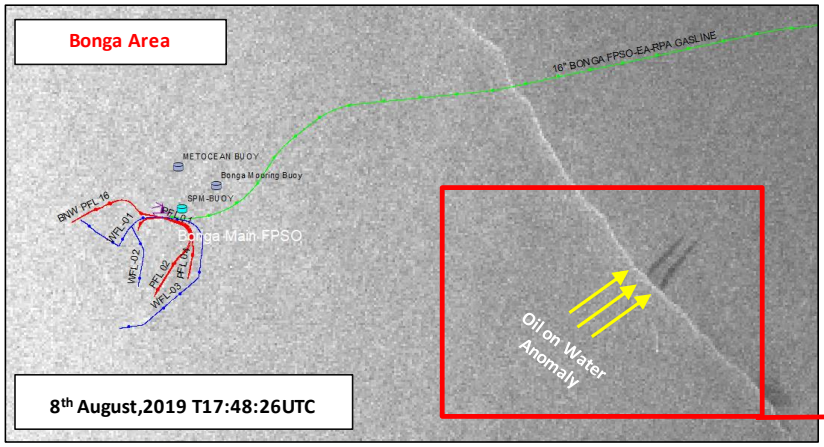
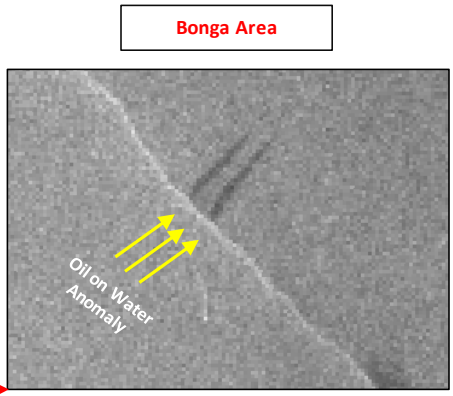


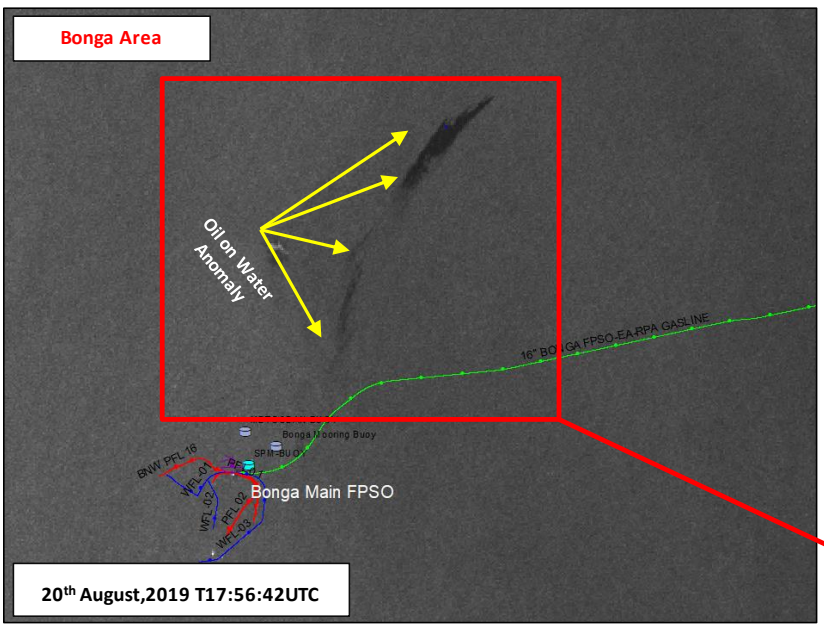
Image 23rd July,2019 T05:35:18UTC

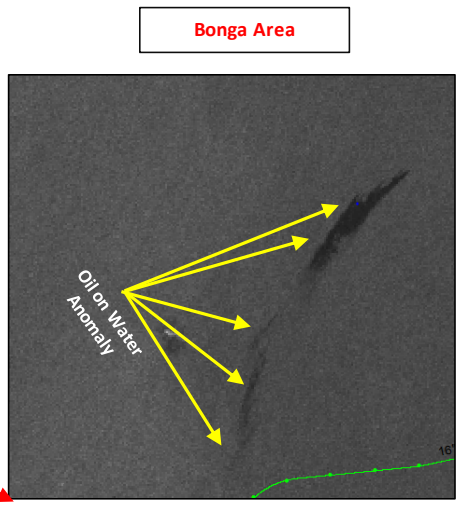
- Oil Analysis from MDA, Radarsat 2, EL1 Beam, HH polarization.A low confidence anomaly identified approximately 9.2km in length in proximity to Bonga FPSO.



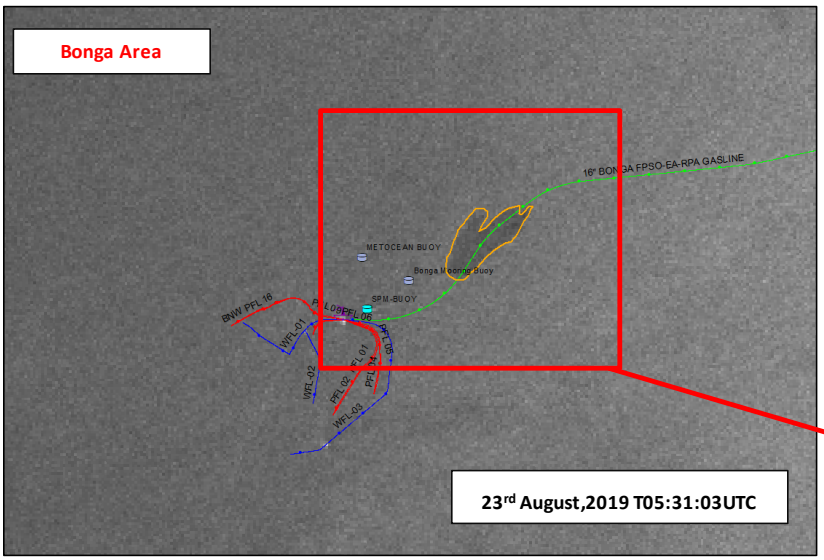


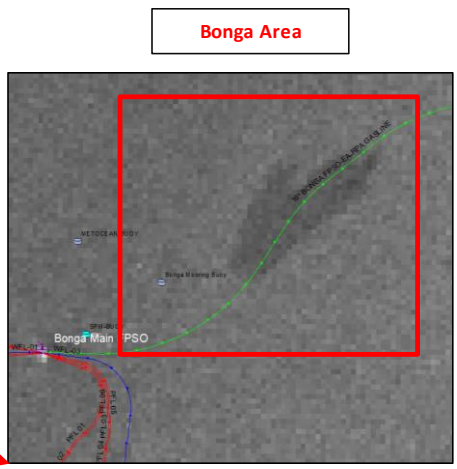
Oil Analysis from MDA, Radarsat 2, SCNA Beam, VV polarization. **8th August,2019 T17:48:26UTC.**





Oil Analysis from MDA, Radarsat 2, SCNB Beam, VV polarization. **20th August,2019 T17:56:42UTC**





Oil Analysis from MDA, Radarsat 2, SCNA Beam, VV polarization. **23rd August,2019 T05:31:03UTC**